Think over Trajectories: Leveraging Video Generation to Reconstruct GPS Trajectories from Cellular Signaling

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Research Questions

- How can video generation models be leveraged to reconstruct high-precision GPS trajectories from coarse cellular signaling data?
- Can trajectory-aware reinforcement learning improve the fidelity of video-generated trajectories under map and temporal constraints?
- Does the Think Over Trajectory paradigm provide better performance than traditional coordinate-based trajectory reconstruction methods?

Core Idea

The paper tackles the Sig2GPS problem: converting coarse cellular tower data into precise GPS trajectories. Traditional approaches use complex multi-stage pipelines, but this work reframes the problem as video generation. The key insight is that domain experts naturally visualize signaling traces on maps and sketch plausible routes - so why not train a model to do the same? They render cellular signaling data as polylines on map images, then train a flow-based video model to progressively draw GPS trajectories over time, mimicking how humans would sketch routes on a map.

Key Formula

Decoupled Group-Relative Advantage

$$A_m^{(i,k)} = \frac{r_m^{(i,k)} - \text{mean}_{k'=1}^{K} r_m^{(i,k')}}{\text{std}_{k'=1}^{K} r_m^{(i,k')} + \epsilon}$$

A_m^{(i,k)} is the normalized advantage for reward component m, trajectory k in group i. r_m^{(i,k)} is the raw reward. This decouples heterogeneous reward scales (distance vs direction vs continuity) by normalizing each component separately within its rollout group before aggregation.

Distance Reward

$$R_{dist} = \frac{1}{|M|} \sum_{m \in M} \left(1 - \frac{\min\{\text{dis}(m), 5000\}}{5000}\right)$$

R_{dist} measures trajectory accuracy at key frames M = {1, ⌊F/4⌋, ⌊F/2⌋, ⌊3F/4⌋, F}. dis(m) is geodesic distance between predicted and ground truth at frame m. The reward is bounded to [0,1] with 5000m maximum penalty, encouraging the generator to stay close to target throughout the drawing process.

Key Figure

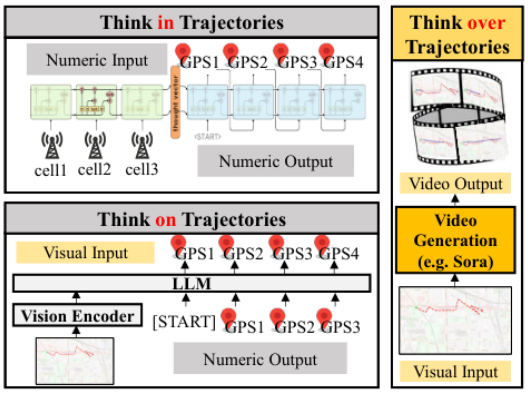

Figure 2: Illustration of Think Over Trajectory. We highlight the difference between previous seq2seq methods, VLM-based methods and our proposed methods.

3 Preliminaries

3.1 Problem Defination

Cellular Signaling Data: In this study, the original cellular signaling data is defined as a temporal sequence of cellular station connections, represented as $T = \{T_1, T_2, \ldots, T_{|T|}\}$, where each element $T_j = \{x_j, y_j, t_j^0, t_j^1\}$ signifies the mobile phone's connection to a cellular station located at (x_j, y_j) from t_j^0 to t_j^1.

GPS Trajectory Data: The ground-truth GPS trajectory is defined as an evenly-sampled sequence $G = \{G_1, G_2, \cdots, G_{|G|}\}$, where each point $G_k = \{\hat{x}_k, \hat{y}_k, \hat{t}_k\}$ denotes the GPS coordinate at timestamp $\hat{t}_k$. The fixed sampling interval is assumed as Δt, i.e., $\hat{t}_{k+1} - \hat{t}_k = \Delta t$.

Sig2GPS Task: Given a cellular signaling sequence T (and the corresponding map context), Sig2GPS aims to reconstruct the corresponding fine-grained GPS trajectory G. Formally, we learn a mapping $\mathcal{F}_\theta$ such that $\hat{G} = \mathcal{F}_\theta(T)$, and $\hat{G}$ is as close as possible to the ground truth G while remaining spatially plausible under map constraints. We align the start time of the GPS trajectory with that of the signaling sequence and resample GPS points at a fixed

Figure 2: Clearly illustrates the key paradigm shift from traditional coordinate-based methods to visual video generation, showing how Think Over Trajectory operates in unified map-visual space rather than abstract coordinate space.

Limitations

- Requires paired signaling-GPS data which may be difficult to obtain due to privacy constraints and cross-system linkage challenges
- Video generation approach is computationally expensive compared to direct coordinate regression methods
- Performance may degrade under sparse signaling conditions or in areas with poor cellular coverage
- Limited evaluation to taxi trajectories which may not represent all mobility patterns (pedestrian, different vehicle types)